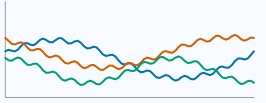
LOMM simulation pipeline

Empirical calibration

A Video data

frames

B FER-like scores



label channels

C Realism targets

- Lag-1 ACF
- Channel IQR
- Derivative IQR
- Run-length entropy
- Spectral entropy
- Effective rank

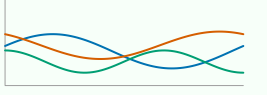
D Scenario selection

- dynamics — η, ζ, σ_q
- activity — dominance, switching, low expression
- measurement — $\Lambda, \Sigma_\varepsilon$

calibrates

Known-truth simulation study

E Latent source truth

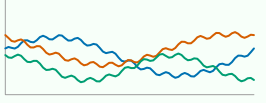


$x(t), v(t)$

F Observation layer

$y = \Lambda \tilde{x} + \varepsilon$

G Simulated panels



known blocks

H Recovery test

$\hat{m}$, blocks

compare estimates with known m and blocks